Spectro-Scout Sample Cases (HO1-HO6) Inputs / Outputs

H01: New API after NIR pilot

Inputs:

Use case description:

Our existing NIR pilot follows API concentration in one powder formulation at the tablet press feed frame. We are introducing a different API and excipient blend and would like to reuse the calibration. A colleague suggests adding Raman so that the old calibration will remain valid. What should our first technical assessment and experiment be?

Target property / analyte:

API concentration in a new powder formulation

Process environment:

Tablet press feed frame; calibration transfer unproven

Measurement goal:

Assess whether the existing calibration remains valid

Reference Method:

Existing NIR pilot and independent API reference assays

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

Why this ranking?

01

Near-infrared (NIR)

First to screen | 780–2500 nm (nominal method range)

Heuristic fit
92/100

NIR diffuse reflection is supported as a candidate for powder-blend concentration. However, a new API/excipient blend is a new matrix: do not assume the old calibration transfers. Rebuild or explicitly validate transfer on the new formulation.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- Adding a second method does not preserve the old NIR calibration automatically.

02

Raman spectroscopy

`Report laser excitation (nm) separately from Raman shift (cm⁻¹)`

Heuristic fit
78/100

Raman is a reasonable independent comparison candidate for the new formulation, consistent with the supplied pharmaceutical examples. Evaluate its own signal and model first.

POTENTIAL STRENGTH

Molecular fingerprint information; supplied pharmaceutical and bioprocess examples support screening it as a candidate alongside NIR.

WATCH-OUTS

Signal may be weak; fluorescence background can overwhelm Raman features. Calibration remains matrix-specific.
Setup to investigate: Fiber Raman probe; screen representative samples for fluorescence and signal quality before choosing excitation/configuration.

- Screen fluorescence background and signal quality; Raman is not a calibration-transfer guarantee.

03

NIR + Raman (combined model)

`780–2500 nm (nominal method range)`

Heuristic fit
68/100

After independent evaluation, compare a combined NIR + Raman model with the best single-method model on held-out batches. Keep the combination only if it improves agreed metrics.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Use the separately evaluated NIR diffuse-reflection and Raman signals; compare combined model with each single-method baseline.

- A combined model is a hypothesis, not a guaranteed gain

Clarifying Questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

What are the old and new API/excipient identities and concentration ranges? `High priority`

A matrix change can alter spectra and calibration validity.

02

What independent API assay, uncertainty and acceptance limits will define success? `High priority`

Calibration transfer needs objective external validation.

03

Which batch, particle-size, density and flow variations must be represented? `Medium priority`

Powder physical variation can confound composition signal.

Feasability plan:

Feasibility plan:
New-formulation calibration-transfer challenge

A low-cost screening experiment designed to retire the biggest technical risks.

`Bench-first`

Proposed protocol

1

Collect representative new-formulation batches spanning the agreed API range and physical variation.

2

Acquire NIR and Raman spectra independently; record process/batch conditions and run independent API assays.

3

Evaluate the old NIR model without refitting as a baseline; quantify bias/error on new material.

4

Build fresh single-method models; only then test combined NIR + Raman against single-method models.

5

Validate on held-out batches or runs, not randomly adjacent spectra from the same batch.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

Pre-agreed error and bias limits are met on independent new-formulation batches.

Combined data demonstrably outperform the best single method if combination is retained.

Calibration scope and transfer limits are documented.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Reference data is essential

Use representative samples and an independent reference method. Avoid claiming quantitative performance from calibration samples alone.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR

Near-infrared

Moisture and organic composition trends

KEY LIMITATION

Broad overlapping overtone/combination bands; calibration required.

MIR / FTIR

Mid-infrared

Functional-group identification and chemical specificity

KEY LIMITATION

Water absorption and optical materials/path length can limit inline use.

Raman

Raman scattering

Molecular fingerprinting, potentially useful in aqueous matrices

KEY LIMITATION

Fluorescence and laser safety must be assessed early.

UV-Vis

Ultraviolet-visible absorption

Colored species, chromophores, and concentration tracking

KEY LIMITATION

Turbidity, bubbles, path length, and fouling can interfere.

Fluorescence

Molecular emission

Highly sensitive measurements for suitable fluorescent targets

KEY LIMITATION

Requires an emitting target or validated marker; quenching may matter.

Decision status: Provisional feasibility assessment

H02: Two Dairy streams

Inputs:

Use case description:

Our dairy process handles both a relatively clear whey stream and an opaque milk concentrate. We want a pilot for composition monitoring, using laboratory protein and fat results where relevant. We can compare different measurement interfaces before choosing installation. What first measurement concepts and comparison experiment make sense?

Target property / Analyte

Protein in whey; protein and fat in opaque concentrate (confirm relevance by stream)

Process environment:

Two distinct optical matrices; interface options can be compared

Measurement goal:

Choose an initial pilot concept for each stream

Reference method:

Laboratory protein and fat results where applicable

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

Why this ranking?

01

Near-infrared (NIR)

First to screen | 780–2500 nm (nominal method range)

Heuristic fit

88/100

Compare two NIR interfaces, not one universal geometry: transmission/transflection for the relatively clear whey; diffuse reflection for the opaque/scattering concentrate. Treat these as separate matrix-specific pilot candidates.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- If whey turbidity changes, test it as a deliberate challenge condition.

02

Mid-infrared (MIR / FTIR)

`4000–400 cm⁻¹ (nominal; approximately 2.5–25 µm)`

Heuristic fit
76/100

MIR ATR is a comparison candidate for the opaque concentrate because ATR is applicable to liquids/pastes in contact with a compatible crystal; verify water absorption, contact and materials.

POTENTIAL STRENGTH

Fundamental vibrational bands can provide chemically informative features; ATR suits liquids, pastes and soft solids in contact with a compatible crystal.

WATCH-OUTS

Usable range depends on fiber, crystal and instrument; water absorption, contact, fouling and chemical compatibility matter.
Setup to investigate: Investigate an ATR interface for compatible liquids/pastes/soft solids; verify the complete optical path and wetted materials with AFS.

- Do not assume it quantifies fat/protein until lab-referenced samples establish performance.

03

Raman spectroscopy

`Report laser excitation (nm) separately from Raman shift (cm⁻¹)`

Heuristic fit
60/100

Optional complementary screen only if NIR/MIR results leave a justified gap; evaluate it independently before any combination.

POTENTIAL STRENGTH

Molecular fingerprint information; supplied pharmaceutical and bioprocess examples support screening it as a candidate alongside NIR.

WATCH-OUTS

Signal may be weak; fluorescence background can overwhelm Raman features. Calibration remains matrix-specific.
Setup to investigate: Fiber Raman probe; screen representative samples for fluorescence and signal quality before choosing excitation/configuration.

- Screen fluorescence and signal quality; do not add a method solely for novelty.

Expert review checkpoint

Confirm target-specific spectral features, process-compatible probe geometry, and any applicable safety or regulatory requirements before proposing hardware.

Clarifiying Questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

Which analytes are required in each stream, and what concentration ranges/limits apply? `High priority`

Protein/fat relevance may differ by stream.

02

What temperature, flow, pressure, fouling and cleaning conditions apply to each interface? High priority

These determine safe installation and repeatable contact/path.

03

What reference assay uncertainty and sampling alignment are available? Medium priority

Spectra and lab samples must represent the same material/time.

Feasibility plan:

Paired two-stream interface comparison

A low-cost screening experiment designed to retire the biggest technical risks.

Bench-first

Proposed protocol

1

Sample whey and opaque concentrate separately across normal and extreme operating states.

2

For whey compare NIR transmission/transflection; include controlled turbidity variation.

3

For concentrate compare NIR diffuse reflection and a compatible MIR ATR concept; record temperature and texture.

4

Pair every spectrum with time-aligned laboratory protein/fat results where relevant.

5

Compare repeatability, bias, prediction error and robustness on held-out runs before choosing an installation.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

Each stream has its own validated method/interface decision.

Turbidity, temperature and texture effects are quantified rather than ignored.

Performance meets agreed limits on independent samples.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Reference data is essential

Use representative samples and an independent reference method. Avoid claiming quantitative performance from calibration samples alone.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR

Near-infrared

Moisture and organic composition trends

KEY LIMITATION

Broad overlapping overtone/combination bands; calibration required.

MIR / FTIR

Mid-infrared

Functional-group identification and chemical specificity

KEY LIMITATION

Water absorption and optical materials/path length can limit inline use.

Raman

Raman scattering

Molecular fingerprinting, potentially useful in aqueous matrices

KEY LIMITATION

Fluorescence and laser safety must be assessed early.

UV-Vis

Ultraviolet-visible absorption

Colored species, chromophores, and concentration tracking

KEY LIMITATION

Turbidity, bubbles, path length, and fouling can interfere.

Fluorescence

Molecular emission

Highly sensitive measurements for suitable fluorescent targets

KEY LIMITATION

Requires an emitting target or validated marker; quenching may matter.

Decision status: Provisional feasibility assessment

HO3: Fluorescence for glucose + viable biomass

INPUTS:

Use case Description:

We have a fluorescence spectrometer and want to report glucose concentration and viable biomass directly during fermentation. Our only justification is that fluorescence can be very sensitive. We have not identified a fluorescent target, marker or reference relationship. What should the assistant tell us?

Target Property/ Analyte:

Glucose concentration and viable biomass

Process environment:

Fluorescence spectrometer available; fluorescent target/marker not identified

Measurement goal:

Direct in-process concentration and viability claims requested

Reference Method:

No reference relationship yet identified

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

`Why this ranking?`

01

Fluorescence / fluorimetry

`First to screen` `Excitation and emission bands depend on the fluorescent species`

Heuristic fit
35/100

Do not recommend fluorescence as a direct glucose or viable-biomass measurement on the evidence given. Sensitivity is not enough: no fluorescent target, marker or validated relationship has been identified.

POTENTIAL STRENGTH

Potentially high sensitivity for a defined fluorescent species or justified indicator.

WATCH-OUTS

High sensitivity does not imply selectivity or direct measurement of every analyte. Quenching and matrix background can mislead.
Setup to investigate: First identify a fluorescent target/marker and establish a reference relationship in the actual matrix.

- No exact sensor configuration is justified yet.

02

Near-infrared (NIR)

`780–2500 nm (nominal method range)`

Heuristic fit
55/100

NIR may be screened as a multivariate composition-trend candidate for glucose only if representative broth samples and reference assays support a model.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- Changing medium and cell population can confound the calibration.

03

UV–Visible spectroscopy

`Target- and instrument-dependent`

Heuristic fit
48/100

An optical-density/UV–Vis measurement can provide a biomass-related optical signal, but it is not by itself a viable-cell count.

POTENTIAL STRENGTH

Can track a defined absorption or optical-density signal when a suitable wavelength/path exists.

WATCH-OUTS

Mixture bands overlap; turbidity and optical density are not equivalent to a specific concentration or viable cell count.
Setup to investigate: Use a defined optical path and target-specific spectral screen; control bubbles, turbidity and fouling.

- Define viability and establish an independent viability reference if that is the required variable.

Expert review checkpoint

Confirm target-specific spectral features, process-compatible probe geometry, and any applicable safety or regulatory requirements before proposing hardware.

Clarifiying Questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

What exactly defines viable biomass, and what independent assay measures it? `High priority`

Optical density/fluorescence cannot be equated with viability without validation.

02

Is there a fluorescent analyte, marker or indicator with a demonstrated relationship to glucose/biomass? `High priority`

Fluorescence is not a universal direct sensor for non-fluorescent targets.

03

Which reference glucose assay and acceptance limits are available? `High priority`

A quantitative claim requires paired reference measurements.

Feasibility plan:

Target-definition and signal feasibility screen

A low-cost screening experiment designed to retire the biggest technical risks.

`Bench-first`

Proposed protocol

1

Separate the requested outputs: glucose concentration versus biomass/viability.

2

Identify candidate fluorescent species/marker and its evidence; if none exists, do not proceed with a direct-fluorescence claim.

3

Collect representative fermentation samples across growth stages and matrix changes.

4

Compare fluorescence, NIR and/or optical-density signals against independent glucose and viability reference assays.

5

Evaluate on held-out runs and document whether each signal tracks the intended target or only a correlated surrogate.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

Each target has an independent reference method and agreed acceptance criteria.

A defined fluorescence marker/relationship is demonstrated before quantitative claims.

Viability is not inferred from optical density alone.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Reference data is essential

Use representative samples and an independent reference method. Avoid claiming quantitative performance from calibration samples alone.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR

Near-infrared

Moisture and organic composition trends

KEY LIMITATION

Broad overlapping overtone/combination bands; calibration required.

MIR / FTIR

Mid-infrared

Functional-group identification and chemical specificity

KEY LIMITATION

Water absorption and optical materials/path length can limit inline use.

Raman

Raman scattering

Molecular fingerprinting, potentially useful in aqueous matrices

KEY LIMITATION

Fluorescence and laser safety must be assessed early.

UV-Vis

Ultraviolet-visible absorption

Colored species, chromophores, and concentration tracking

KEY LIMITATION

Turbidity, bubbles, path length, and fouling can interfere.

Fluorescence

Molecular emission

Highly sensitive measurements for suitable fluorescent targets

KEY LIMITATION

Requires an emitting target or validated marker; quenching may matter.

Decision status: Insufficient information

HO4: MIR ATR Petroleum + silica link

INPUTS:

Use case description:

We want to investigate water content in a petroleum line using MIR ATR. The analyzer would be in another building. We have a silica fiber cable from a visible or NIR setup and want to reuse it. Link length, routing and operating conditions are unknown. Can the concept be approved?

Target property/ analyte:

Water content in petroleum using MIR ATR

Process environment:

Analyzer in another building; existing visible/NIR silica link; length/routing/conditions unknown

Measurement goal:

Assess optical link feasibility

Reference Method:

Laboratory water determination (to be confirmed)

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

Why this ranking?

01

Mid-infrared (MIR / FTIR)

First to screen | 4000–400 cm^{-1} (nominal; approximately 2.5–25 µm)

Heuristic fit

88/100

MIR ATR is a relevant concept to investigate for water in petroleum, but the existing visible/NIR silica cable must not be assumed suitable for MIR. The supplied fiber table lists silica families only to about 1200 nm or 2400 nm; MIR needs a compatible IR path.

POTENTIAL STRENGTH

Fundamental vibrational bands can provide chemically informative features; ATR suits liquids, pastes and soft solids in contact with a compatible crystal.

WATCH-OUTS

Usable range depends on fiber, crystal and instrument; water absorption, contact, fouling and chemical compatibility matter.
Setup to investigate: Investigate an ATR interface for compatible liquids/pastes/soft solids; verify the complete optical path and wetted materials with AFS.

- A complete link assessment is required: fiber family, length, bends/routing, connectors, optical losses, temperature, pressure and hazardous-area constraints.

02

Near-infrared (NIR)

780–2500 nm (nominal method range)

Heuristic fit

52/100

NIR is a different spectral approach and could be a separate alternative study if the target signal and matrix support it; it does not make a visible/NIR silica cable into a MIR link.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- Do not conflate NIR and MIR wavelength ranges.

Clarifying Questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

What exact fiber type, length, routing/bend radius and connectors are installed? High priority

Link transmission and losses are configuration-specific.

02

What are process temperature, pressure, line materials and hazardous-area requirements? High priority

These govern probe and installation suitability.

03

Which MIR spectrometer, ATR crystal and required spectral range are proposed? High priority

All components must share a compatible usable range.

Feasibility plan:

Optical-link and interface feasibility gate

A low-cost screening experiment designed to retire the biggest technical risks.

Bench-first

Proposed protocol

1

Do not approve reuse of the existing silica cable for MIR ATR based on its visible/NIR use alone.

2

Ask AFS to verify transmission bands for the exact fiber, probe crystal, connectors and analyzer.

3

Document cable length, bends, routing, environmental exposure and process interface constraints.

4

Bench-test optical throughput and spectral quality across the required range using the proposed complete link.

5

Only then test petroleum samples against the independent lab water method.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

AFS confirms compatible transmission for every optical component across the required MIR range.

Signal quality remains sufficient through the actual link length/routing.

Process and hazardous-area requirements are reviewed before installation.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Reference data is essential

Use representative samples and an independent reference method. Avoid claiming quantitative performance from calibration samples alone.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR

Near-infrared

Moisture and organic composition trends

KEY LIMITATION

Broad overlapping overtone/combination bands; calibration required.

MIR / FTIR

Mid-infrared

Functional-group identification and chemical specificity

KEY LIMITATION

Water absorption and optical materials/path length can limit inline use.

Raman

Raman scattering

Molecular fingerprinting, potentially useful in aqueous matrices

KEY LIMITATION

Fluorescence and laser safety must be assessed early.

UV-Vis

Ultraviolet-visible absorption

Colored species, chromophores, and concentration tracking

KEY LIMITATION

Turbidity, bubbles, path length, and fouling can interfere.

Fluorescence

Molecular emission

Highly sensitive measurements for suitable fluorescent targets

KEY LIMITATION

Requires an emitting target or validated marker; quenching may matter.

Decision status: Insufficient information — do not approve as specified

HO5: Unknown off-spec liquid

INPUTS:

Use case description:

A liquid process occasionally goes off specification. We suspect contamination but cannot name the contaminant or affected quality variable. We want the best spectroscopy method and exact sensor configuration immediately. What information and first experiment should come before that decision?

Target Property/ Analyte:

Unknown contamination / off-spec quality variable

Process Environment:

Process conditions and interface unknown

Measurement goal:

Detect off-spec events; detection criteria not defined

Reference method:

Not specified

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

Why this ranking?

01

Near-infrared (NIR)

First to screen | 780–2500 nm (nominal method range)

Heuristic fit

42/100

A broad screening candidate only after the off-spec variable and sample matrix are characterized; no defensible “best method” or exact sensor configuration can be selected from the current facts.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- Unknown contaminant and unknown quality variable prevent target-specific selection.

02

Mid-infrared (MIR / FTIR)

4000–400 cm^{-1} (nominal; approximately 2.5–25 µm)

Heuristic fit

42/100

MIR may be useful if chemical identity/functional-group information is needed and a compatible interface is available, but it is not yet rankable against other methods.

POTENTIAL STRENGTH

Fundamental vibrational bands can provide chemically informative features; ATR suits liquids, pastes and soft solids in contact with a compatible crystal.

WATCH-OUTS

Usable range depends on fiber, crystal and instrument; water absorption, contact, fouling and chemical compatibility matter.
Setup to investigate: Investigate an ATR interface for compatible liquids/pastes/soft solids; verify the complete optical path and wetted materials with AFS.

- Water absorption and interface compatibility may be decisive.

03

Raman spectroscopy

`Report laser excitation (nm) separately from Raman shift (cm⁻¹)`

Heuristic fit

38/100

Raman may provide a molecular fingerprint for some liquids, but target signal and fluorescence background must first be established.

POTENTIAL STRENGTH

Molecular fingerprint information; supplied pharmaceutical and bioprocess examples support screening it as a candidate alongside NIR.

WATCH-OUTS

Signal may be weak; fluorescence background can overwhelm Raman features. Calibration remains matrix-specific.
Setup to investigate: Fiber Raman probe; screen representative samples for fluorescence and signal quality before choosing excitation/configuration.

- No claim that it can identify an unspecified contaminant.

Clarifying questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

What quality variable goes off specification, and what limits define an event? `High priority`

The required measurement target must be defined first.

02

What contaminants are plausible, and what are the normal/off-spec sample compositions? `High priority`

Technique selection depends on target chemistry and spectral contrast.

03

What independent test can label representative normal and off-spec samples? `High priority`

A first screen needs defensible reference labels/results.

04

What are the process conditions, access, cleaning and response-time requirements? `Medium priority`

These determine interface feasibility.

Feasability plan:

Problem definition before sensor selection

A low-cost screening experiment designed to retire the biggest technical risks.

`Bench-first`

Proposed protocol

1

Collect normal and off-spec examples; document process state and the suspected cause without treating suspicion as fact.

2

Define the affected quality variable and obtain independent lab analysis for each sample.

3

Run a broad laboratory spectral screen across representative samples using accessible candidate methods/interfaces.

4

Look for repeatable spectral differences linked to the reference variable; separate nuisance changes such as temperature and turbidity.

5

Only rank methods and specify a sensor after the target signal, interface and success criteria are established.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

A measurable target and independent reference labels are defined.

Candidate signal distinguishes relevant off-spec cases from normal variability on held-out samples.

No exact configuration is claimed before compatibility and sample evidence exist.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR

Near-infrared

Moisture and organic composition trends

KEY LIMITATION

Broad overlapping overtone/combination bands; calibration required.

MIR / FTIR

Mid-infrared

Functional-group identification and chemical specificity

KEY LIMITATION

Water absorption and optical materials/path length can limit inline use.

Raman

Raman scattering

Molecular fingerprinting, potentially useful in aqueous matrices

KEY LIMITATION

Fluorescence and laser safety must be assessed early.

UV-Vis

Ultraviolet-visible absorption

Colored species, chromophores, and concentration tracking

KEY LIMITATION

Turbidity, bubbles, path length, and fouling can interfere.

Fluorescence

Molecular emission

Highly sensitive measurements for suitable fluorescent targets

KEY LIMITATION

Requires an emitting target or validated marker; quenching may matter.

Decision status: Insufficient information

HO6: Kidney study → live surgery

INPUTS:

Use case description:

We read the supplied LED NIR kidney study and want to use the same concept to guide tumor-boundary decisions during live surgery. Can we quote its findings as proof that the sensor is ready? We have no new in-vivo data and no validation on our own setup.

Target Property / Analyte:

Tumor boundary decisions during live surgery

Process environment:

Clinical use; no in-vivo data or own-setup validation

Measurement goal:

Guide surgical decisions

Reference method:

No validation data for proposed setup

OUTPUTS:

Recommendations:

Promising spectroscopy approaches

Ranked heuristic candidates—not a final instrument specification.

Why this ranking?

01

Near-infrared (NIR)

First to screen | 780–2500 nm (nominal method range)

Heuristic fit

25/100

The supplied LED NIR kidney study is a research proof of concept using 14 in-vitro clinical samples (seven tumor/normal pairs from seven patients). It is not evidence of readiness for live-surgery tumor-boundary decisions.

POTENTIAL STRENGTH

Useful for rapid composition trends in powders and many process liquids; fiber-probe geometries include diffuse reflection, transmission and transflection.

WATCH-OUTS

Broad overlapping bands; scattering, temperature and matrix changes can shift the signal. Calibration is specific to the formulation/matrix.
Setup to investigate: Select diffuse reflection for scattering powders/opaque materials; transmission or transflection for clearer liquids. Confirm actual system range.

- Do not quote it as clinical validation or claim performance on a new sensor/setup.

02

Mid-infrared (MIR / FTIR)

4000–400 cm^{-1} (nominal; approximately 2.5–25 µm)

Heuristic fit

20/100

MIR/ATR and fluorescence research can motivate further study, but the supplied model-tissue work does not establish safety or efficacy for live surgery.

POTENTIAL STRENGTH

Fundamental vibrational bands can provide chemically informative features; ATR suits liquids, pastes and soft solids in contact with a compatible crystal.

WATCH-OUTS

Usable range depends on fiber, crystal and instrument; water absorption, contact, fouling and chemical compatibility matter.

Setup to investigate: Investigate an ATR interface for compatible liquids/pastes/soft solids; verify the complete optical path and wetted materials with AFS.

- Clinical translation requires independent in-vivo and workflow validation, regulatory/ethics review and expert clinical oversight.

Clarifying Questions:

What we still need to know

Resolve high-impact unknowns before committing to a method.

01

What clinical protocol, intended-use claim and decision threshold are proposed? `High priority`

A research classification result is not an approved clinical decision aid.

02

What prospective in-vivo validation and independent patient cohort will be used? `High priority`

The supplied study used a small ex-vivo/in-vitro sample set.

03

What safety, sterility, human-factors, regulatory and ethics approvals are required? `High priority`

Surgical use creates requirements beyond optical feasibility.

Feasibility plan:

Research-to-clinical translation gate

A low-cost screening experiment designed to retire the biggest technical risks.

`Bench-first`

Proposed protocol

1

Describe the paper accurately: two sensor prototypes tested on 14 in-vitro clinical samples; do not generalize to live surgery.

2

Reproduce the measurement on the intended hardware and representative ex-vivo samples as a technical feasibility study.

3

Predefine endpoints and validate on an independent patient-level dataset to avoid sample leakage.

4

Only after technical evidence, develop a prospective, ethically approved in-vivo validation plan with clinical, regulatory and safety experts.

5

Do not use the prototype to guide care until the applicable validation and authorization requirements are met.

Define success before testing

Agree quantitative thresholds with the customer and AFS expert.

Study claims are limited to the actual in-vitro sample set and protocol.

Independent validation demonstrates performance on intended hardware and population.

Clinical use is gated by ethics, safety, regulatory and prospective validation—not by a proof-of-concept paper.

Decision gate

If the target signal is weak, confounded, or unstable, revisit technique, probe geometry, and sample presentation before building a calibration model.

Method Guide:

Quick method guide

Use these as screening heuristics; target-specific spectra and process constraints decide feasibility.

NIR
Near-infrared
Moisture and organic composition trends
KEY LIMITATION
Broad overlapping overtone/combination bands; calibration required.
MIR / FTIR
Mid-infrared
Functional-group identification and chemical specificity
KEY LIMITATION
Water absorption and optical materials/path length can limit inline use.
Raman
Raman scattering
Molecular fingerprinting, potentially useful in aqueous matrices
KEY LIMITATION
Fluorescence and laser safety must be assessed early.
UV-Vis
Ultraviolet-visible absorption
Colored species, chromophores, and concentration tracking
KEY LIMITATION
Turbidity, bubbles, path length, and fouling can interfere.
Fluorescence
Molecular emission
Highly sensitive measurements for suitable fluorescent targets
KEY LIMITATION
Requires an emitting target or validated marker; quenching may matter.

Decision status: Outside supported scope for clinical readiness claim